

Portfolio Optimization

Vicente Alfaro

Let n be the number of assets available for investment, $w \in \mathbb{R}^n$ be the vector of weights of the assets in the portfolio, $\mu \in \mathbb{R}^n$ be the vector of expected returns of the assets, and $\Sigma \in \mathbb{R}^{n \times n}$ be the covariance matrix of the asset returns. Under the mean-variance framework, there is not only one optimal portfolio, but rather a set of portfolios known as the efficient frontier. The efficient frontier is a curve that represents the set of portfolios that offer the highest expected return for a given level of risk (measured by the portfolio's variance) or, equivalently, the lowest risk for a given level of expected return. Let's review some of the most common portfolio optimization approaches:

1 Minimum Variance

The minimum variance portfolio is the portfolio that has the lowest possible risk (variance) among all possible portfolios. The optimization problem can be formulated as follows:

$$\begin{aligned} \min_w \quad & w^\top \Sigma w \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1 \end{aligned}$$

Also, we can add a constraint to ensure that the expected return of the portfolio is equal to a target return R :

$$\mu^\top w = R$$

and a constraint to ensure that the weights are non-negative (no short selling):

$$w \geq 0$$

Note that using an equality constraint traces out the *entire* hyperbola as R varies, including the inefficient (dominated) lower branch for $R < \mu_{GMV}$ — the dashed branch shown in Figure 1. Replacing it with $\mu^\top w \geq R$ instead would restrict the solution to the efficient branch only, coinciding with the global minimum-variance portfolio whenever $R \leq \mu_{GMV}$.

2 Risk Aversion Formulation

The risk aversion formulation is a way to balance the trade-off between expected return and risk. The optimization problem can be formulated as follows:

$$\begin{aligned} \max_w \quad & \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1 \end{aligned}$$

where $\lambda \geq 0$ is the risk aversion coefficient. When λ is small, the penalty from the contribution of the portfolio risk is also small, leading to more risky portfolios. Conversely, when λ is large,

portfolios with more exposure to risk become more highly penalized. It is also possible to add a no short selling constraint to this formulation.

3 Risk-free Asset

Until now, we have assumed that every available asset has risk $\sigma_i > 0$ for $i \in [n]$. Now, assume that we can also invest in one risk-free asset (e.g., a government bond) yielding a return r_f . Under this condition, the efficient frontier is no longer a curve, but rather a straight line originating from $(0, r_f)$ that is tangent to the risky-asset efficient frontier at a single point. This point is called the tangency portfolio (with weights w_T). To find the weights of the tangency portfolio w_T , we must find the portfolio that maximizes the Sharpe Ratio, which measures the expected excess return per unit of risk. The optimization problem is inherently fractional:

$$\begin{aligned} \max_w \quad & \frac{\mu^\top w - r_f}{\sqrt{w^\top \Sigma w}} \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1 \\ & w \geq 0 \quad (\text{optional}) \quad (*) \end{aligned}$$

Because standard solvers like Gurobi are designed to handle Linear (LP) or Quadratic Programming (QP) problems, they cannot solve this fractional objective directly. However, we can linearize the problem by applying the Charnes-Cooper transformation. The transformed QP problem becomes::

$$\begin{aligned} \min_y \quad & y^\top \Sigma y \\ \text{s.t.} \quad & (\mu - r_f \mathbf{1})^\top y = 1 \\ & y \geq 0 \quad (\text{iff we added } (*)) \end{aligned}$$

After solving this standard QP problem for the optimal auxiliary variables y^* , the actual weights of the tangency portfolio w_T are recovered by normalizing y^* so that they sum to one:

$$w_T = \frac{y^*}{\mathbf{1}^\top y^*}$$

With this portfolio w_T , the efficient frontier becomes the Capital Market Line (CML), representing combinations of the risk-free asset and the tangency portfolio.

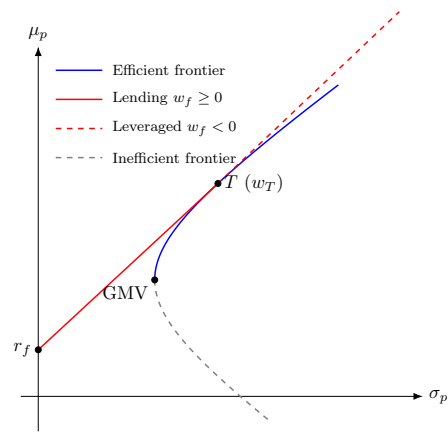


Figure 1: Representation of the efficient frontier with a risk free asset.